

Practical: Jenkins Automated Build Trigger using GitHub Webhook

Aim

To configure Jenkins to automatically trigger and execute a Python program when new code is pushed to a GitHub repository using GitHub Webhook.

Requirements

Jenkins installed and running

Git installed

GitHub account

Python installed

Internet connection

Step 1: Start Jenkins

Open Terminal and run:

```
sudo systemctl start jenkins.service  
sudo systemctl status jenkins.service
```

Open browser:

<http://localhost:8080>

Login using:

Username: admin

Password: admin123

Step 2: Install Required Plugins

Go to:

Manage Jenkins → Manage Plugins → Available Plugins

Install:

Git Plugin

Restart Jenkins.

Step 3: Create GitHub Repository

Go to GitHub.

Click:

New Repository → Enter name → Click Create Repository

Create a file named:

`add.py`

Example code:

```
a = 10
b = 20
print("Addition is:", a+b)
```

Commit and push the file.

Step 4: Create New Job in Jenkins

Go to Jenkins Dashboard.

Click:

New Item

Enter:

`Addition_Project`

Select:

Freestyle Project

Click OK.

Step 5: Connect Jenkins with GitHub Repository

Click Configure.

Go to Source Code Management.

Select Git.

Enter Repository URL:

`https://github.com/yourusername/yourrepository.git`

In Branches to build, enter:

`*/main`

Click Save.

Step 6: Add Build Step

Click Configure.

Scroll to Build section.

Click Add Build Step → Execute Shell

Enter:

```
echo "Build Started"
python3 add.py
echo "Build Finished"
```

Click Save.

Step 7: Enable GitHub Webhook Trigger in Jenkins

Click Configure.

Scroll to Build Triggers.

Select:

`GitHub hook trigger for GITScm polling`

Click Save.

Step 8: Find System IP Address

Open Terminal and run:

`ip addr`

Find your IP address.

Example:

`172.16.3.199`

Your Jenkins webhook URL will be:

`http://172.16.3.199:8080/github-webhook/`

Step 9: Configure Webhook in GitHub

Go to GitHub repository.

Click:

Settings → Webhooks → Add Webhook

Enter:

Payload URL:

`http://172.16.3.199:8080/github-webhook/`

Content type:

`application/json`

Select:

`Just the push event`

Click:

Add Webhook

Step 10: Test Automatic Build

Make change in add.py file.

Example:

```
print("Webhook Triggered Successfully")
```

Push code using terminal:

```
git add .  
git commit -m "Webhook test"  
git push origin main
```

Step 11: Verify Build in Jenkins

Go to Jenkins Dashboard.

Click your job.

Click Build History → Latest Build → Console Output

Output will show:

```
Build Started  
Addition is: 30  
Webhook Triggered Successfully  
Build Finished  
Finished: SUCCESS
```

Result

Jenkins successfully triggered automatic build and executed the Python program when code was pushed to GitHub using GitHub Webhook.

Workflow Diagram (Text Explanation)

Developer → Push Code → GitHub → Webhook → Jenkins → Build Triggered → Execute add.py → Output Generated

Conclusion

Jenkins was successfully integrated with GitHub using webhook, enabling automatic build execution whenever new code was pushed to the repository.